

Fayetteville, AR
(479) 463-0914
ark0723@gmail.com

Ahla Ko

github.com/ark0723
linkedin.com/in/ahlako

Summary

Research Engineer specializing in machine learning, reinforcement learning, and computer vision, with extensive experience developing AI-driven healthcare applications. Proven ability to take ML systems from research prototype to production using Python, PyTorch, FastAPI, and modern AI frameworks.

Skills

- Language: Python
- Front-end: Pytorch, Tensorflow, Numpy, Pandas, OpenCV, Databricks
- Back-end: SQL, FastAPI, Django, Git, Docker, AWS S3, AWS EC2

Experience

Research Engineer Healthcare AI Research	University of Arkansas Fayetteville, AR	Aug 2018 – 2026
--	---	------------------------

- Designed and developed an end-to-end reinforcement learning–based clinical decision support system that personalized breast cancer prevention strategies for high-risk patients, improving outcomes by up to 4.33 QALYs over standard clinical guidelines using PPO and Markov Decision Processes.
- Engineered an end-to-end medical document digitization pipeline using YOLOv11, Residual CRNN, and OpenCV, achieving 99.67% OCR accuracy, automating 91% of manual data entry, and reducing processing costs by over 90% through human-in-the-loop validation.
- Designed and implemented a custom OpenAI Gymnasium environment for metastatic breast cancer treatment optimization, combining reinforcement learning, sensitivity analysis, and SHAP interpretability to identify patient-specific treatment strategies and clinically explainable recommendations.

Back-End Developer Early Team	Heroworks Busan, Korea	2021
---	----------------------------------	-------------

- Developed backend services and REST APIs for a hotel revenue management platform supporting pricing intelligence across 20+ OTA channels using Flask, SQLAlchemy, and PostgreSQL.
- Collaborated with the Lead Developer to design relational database schemas and data models supporting hotel pricing, demand forecasting, and competitive market analysis.
- Built data-driven monitoring tools including real-time pricing alerts, interactive analytics dashboards, and automated reporting workflows that enabled hotel operators to monitor pricing trends and make revenue optimization decisions

Research Analyst Quantitative Analysis Team	Korea Institute for Defense Analysis Seoul, Korea	2016 – 2017
---	---	--------------------

- Designed mathematical optimization models for military facility location planning, minimizing land acquisition costs while satisfying operational and environmental constraints.
- Synthesized multi-agency defense datasets and delivered quantitative analyses supporting strategic planning and Korea–U.S. defense cost-sharing negotiations.

Projects

Signal Matrix

2026 – Present

- Designed an automated equity screening platform using FastAPI, vectorbt, PostgreSQL, and GitHub Actions, supporting daily screening across 500+ S&P 500 stocks.
- Integrated technical, momentum, fundamental, and macroeconomic signals (FRED) into a unified screening engine with automated Telegram notifications.
- Built a production-ready investment research workflow with Clerk-authenticated APIs, scheduled data pipelines, and scalable backend services.

Real-time Anomaly Detection

2025

- **Developed** image anomaly detection models on **Azure Databricks** with MLflow experiment tracking and model versioning.
- **Migrated** cloud inference to a lightweight **FastAPI + Streamlit** deployment, enabling real-time local image prediction while simplifying infrastructure and reducing deployment complexity.

Binance MCP Server

2025

- Implemented an MCP-compatible cryptocurrency data server using FastMCP, exposing real-time market data and analytics tools for AI agents.

AI-Powered Custom Apparel Platform

2024

- Built backend services for an AI-powered apparel platform using FastAPI/Django microservices, Docker, JWT authentication, Nginx, and AWS S3.

Education

University of Arkansas

Fayetteville, AR

2026

- Ph.D. in Industrial Engineering
- Dissertation: Deep Learning and Decision Analytics in Healthcare Systems
- Teaching Assistant (INEG 2313 I & II): Mentored undergraduate students in Applied Probability & Statistics across multiple semesters.

Pusan National University

Busan, Korea

- M.S. in International Logistics and Port Management
- Thesis: Optimization of Blood Platelet Production and Inventory Management
- B.S. in Chemistry Education

2015

2011

Publications

Deep Learning-Based Automated Recognition of Hand-Filled Forms for Medical Study Questionnaires in Moldova

Ahla Ko, Yanjun Pan, Donald Catanzaro, Valeriu Crudu, Shengfan Zhang. *IEEE Transactions on Industrial Informatics* (Under Review)